INTAKE 46

Windows Server

>_ System Administration — ITI Alexandria

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AD DS — GPO — IIS/FTP — DNS/DHCP — WDS

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Contents

1 Introduction **3**
1.1 Project Objectives . . . 3
1.2 Network Topology . . . 4
1.2.1 Network Configuration Summary . . . 4

2 Active Directory Forest and Domain Hierarchy **5**
2.1 Parent Domain: ITI.LOCAL . . . 5
2.2 Child Domain: ALEX.ITI.LOCAL . . . 5
2.3 Child Domain: ISM.ITI.LOCAL . . . 6

3 Implementing Login Restrictions for A@ITI.local **6**
3.1 Restricting Login to PC1 Only . . . 7
3.2 Configuring Time-Based Restrictions (No Login on Fridays) . . . 7
3.3 Verification . . . 8

4 Configuring RODC Access for help@ITI.local **8**
4.1 Password Replication Policy Configuration . . . 8
4.2 Verification of RODC Authentication . . . 9

5 Implementing Restrictions for c@ITI.local **9**
5.1 Configuration . . . 9
5.1.1 Removing Access to Control Panel . . . 9
5.1.2 Blocking Removable Storage and Flash Memory Access . . . 9
5.1.3 Enforcing Desktop Wallpaper (ITI Logo) . . . 10
5.2 Verification . . . 10
5.2.1 Verifying Control Panel Restriction . . . 10
5.2.2 Verifying Removable Storage Restriction . . . 11
5.2.3 Verifying Desktop Wallpaper Enforcement . . . 11

6 Software Deployment Using Group Policy Objects (GPO) **12**
6.1 GPO Configuration for VLC Installation . . . 12
6.1.1 Creating Software Deployment GPO . . . 12
6.1.2 Configuring Startup Policy . . . 12
6.2 Verification of VLC Installation . . . 13

7 Administrative Delegation for B@iti.local **13**
7.1 Configuration . . . 14
7.1.1 Adding B@iti.local to Remote Desktop Users Group . . . 14
7.1.2 Enabling Remote Desktop via Group Policy . . . 14
7.2 Verification of Delegated Access . . . 15

8 Roaming User Profiles for A@Ism.ITI.Local **15**
8.1 Configuration . . . 16
8.1.1 Creating Shared Folder for Profiles . . . 16
8.1.2 Configuring User Profile Path . . . 16
8.2 Verification of Roaming Profile . . . 17

9 Remote Server Configuration (External to Domain) 17
9.1 Overview . . . 17
9.2 Web Server Configuration . . . 18
9.3 DNS Configuration . . . 18
9.3.1 Method 1: Local DNS Service on Remote Server . . . 18
9.3.2 Method 2: Secondary Zone on Primary Domain Controller . . . 19
9.3.3 Verification of DNS Configuration and Secondary Zone . . . 19
9.4 Installing DHCP Server Role . . . 20
9.4.1 Creating DHCP Scope . . . 20
9.4.2 Verification of DHCP Configuration . . . 21
9.5 Remote Desktop Protocol (RDP) Configuration . . . 21
9.6 FTP Server Configuration . . . 22
9.6.1 Verification of FTP Access . . . 22
9.6.2 Verification: User Can Create Own Folder . . . 22
9.6.3 Verification: User Cannot Access Other Users' Folders . . . 23

10 Windows Deployment Services (WDS) for Automated Deployment 23
10.1 Creating 50 User Accounts Using Batch Script . . . 23
10.1.1 CMD Batch Script for Bulk User Creation . . . 23
10.1.2 Verification of User Creation . . . 24
10.2 WDS Server Configuration on DC6 . . . 24
10.2.1 Installing WDS Server Role . . . 24
10.2.2 Configuring DHCP Scope for WDS . . . 24
10.2.3 Adding Boot Image (boot.wim) . . . 25
10.2.4 Adding Install Image (install.wim) . . . 26
10.3 Testing WDS Deployment . . . 26
10.3.1 PXE Boot Process . . . 26
10.3.2 Windows Installation via WDS . . . 27
10.3.3 Successful Deployment Verification . . . 27

11 Conclusion 28

1 Introduction

Active Directory (AD) is Microsoft's directory service for Windows networks that provides centralized management of users, computers, and resources. It enables organizations to control authentication, enforce security policies, and manage IT infrastructure from a single location.
In enterprise environments, Active Directory Domain Services (AD DS) is essential for managing large-scale networks efficiently. A well-designed AD infrastructure with multiple domain controllers provides redundancy, improves performance, and enhances security.
This project implements a Windows Server Active Directory forest that simulates a real-world enterprise environment. The implementation includes multiple domains, Read-Only Domain Controllers (RODC), Group Policy Objects (GPO), roaming profiles, administrative delegation, and automated deployment services to demonstrate comprehensive Active Directory capabilities.

1.1 Project Objectives

The primary objectives of this project are:

1. **Design and implement a multi-domain Active Directory forest** consisting of ITI.Local as the root domain with two child domains (Alex.ITI.Local and Ism.ITI.Local).
2. **Deploy multiple domain controllers** with different roles:
   - DC1: Primary Domain Controller with DNS services
   - DC2: Additional Domain Controller for redundancy
   - DC3: Read-Only Domain Controller (RODC)
   - DC4 & DC5: Domain Controllers for child domains
   - DC6: Windows Deployment Services (WDS) server
3. **Implement user access control policies** including computer-specific login restrictions, time-based access control, and system resource restrictions.
4. **Configure RODC authentication** for branch office scenarios with password replication policies.
5. **Deploy software using Group Policy Objects (GPO)** to automate application installation across the domain.
6. **Implement administrative delegation** to grant specific users limited administrative rights without full domain administrator privileges.
7. **Configure roaming user profiles** to enable seamless user experience across multiple workstations.
8. **Automate workstation deployment** using Windows Deployment Services to efficiently provision multiple computers and user accounts.
9. **Set up external network services** including DNS, DHCP, and Web Server infrastructure.
10. **Configure remote access and file transfer services** with appropriate security controls.

1.2 Network Topology

Figure 1 illustrates the complete network infrastructure implemented in this project.

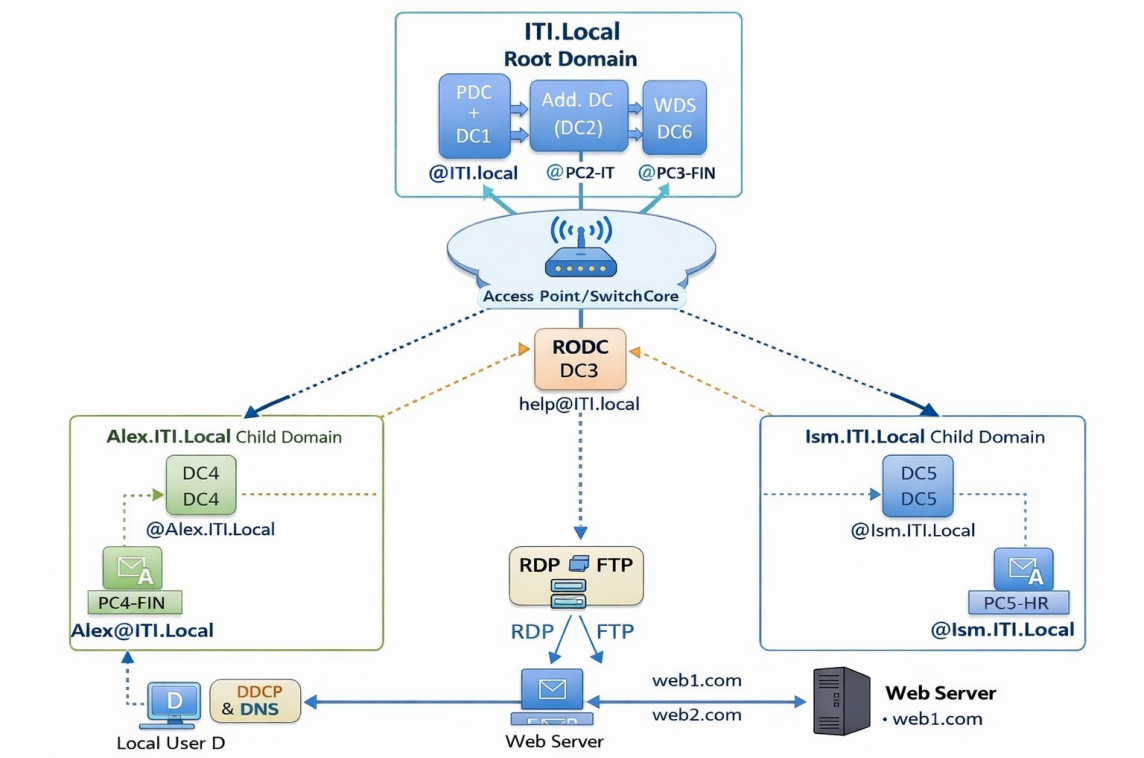


Figure 1: Network Topology Diagram showing ITI.Local forest with child domains, domain controllers, client workstations, and external services

1.2.1 Network Configuration Summary

Device	IP Address	Role/Domain
DC1	10.166.108.1	Primary DC + DNS (ITI.Local)
DC2	10.166.108.2	Additional DC (ITI.Local)
DC3	10.166.108.3	RODC (ITI.Local)
DC4	10.166.108.4	DC (Alex.ITI.Local)
DC5	10.166.108.5	DC (Ism.ITI.Local)
DC6	10.166.108.6	WDS Server (ITI.Local)
Web/FTP Server	10.166.108.47	External Server
PC6	10.166.108.50	Special Workstation
PC1-HR	10.166.108.51	ITI.Local Domain
PC2-IT	10.166.108.52	ITI.Local Domain
PC3-FIN	10.166.108.53	ITI.Local Domain
PC4-FIN	10.166.108.54	Alex.ITI.Local Domain
PC5-HR	10.166.108.55	Ism.ITI.Local Domain

Table 1: IP Address Assignment Table

Network Parameters:

- **Network Address:** 10.166.108.0

- **Subnet Mask:** 255.0.0.0
- **Default Gateway:** 10.166.108.254 (Mobile Hotspot)
- **Primary DNS Server:** 10.166.108.1 (DC1)

2 Active Directory Forest and Domain Hierarchy

The Active Directory infrastructure consists of one forest root domain (ITI.Local) and two child domains (Alex.ITI.Local and Ism.ITI.Local). Each domain contains multiple Organizational Units (OUs) to organize users, computers, and resources based on departmental structure.

2.1 Parent Domain: ITI.LOCAL

The root domain ITI.LOCAL contains the primary domain controllers (DC1, DC2, and DC3-RODC) and multiple organizational units for different departments including HR, IT, FIN, and Sales.

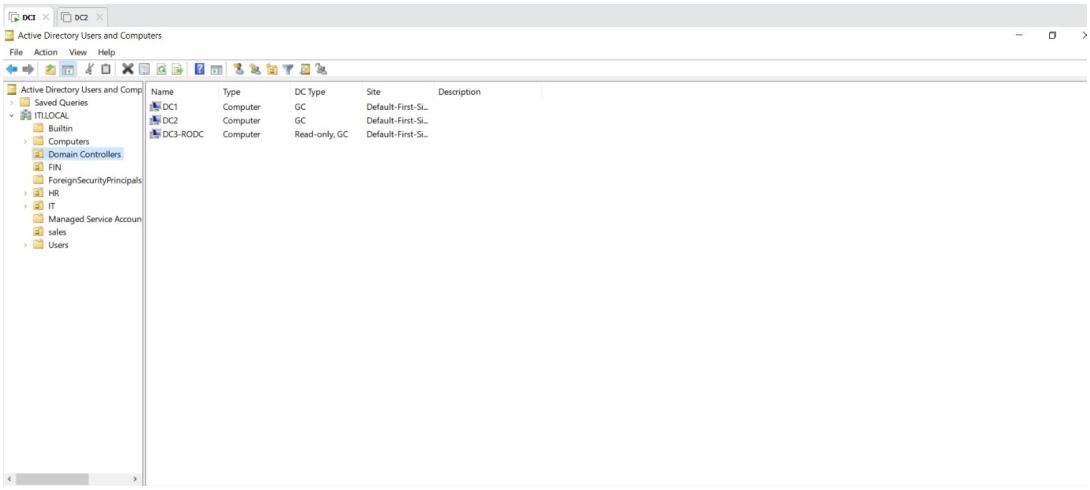


Figure 2: ITI.LOCAL parent domain structure showing Domain Controllers (DC1, DC2, DC3-RODC) and Organizational Units (HR, IT, FIN, Sales)

2.2 Child Domain: ALEX.ITI.LOCAL

The ALEX.ITI.LOCAL child domain contains DC4 as the domain controller and organizational units including the FIN department.

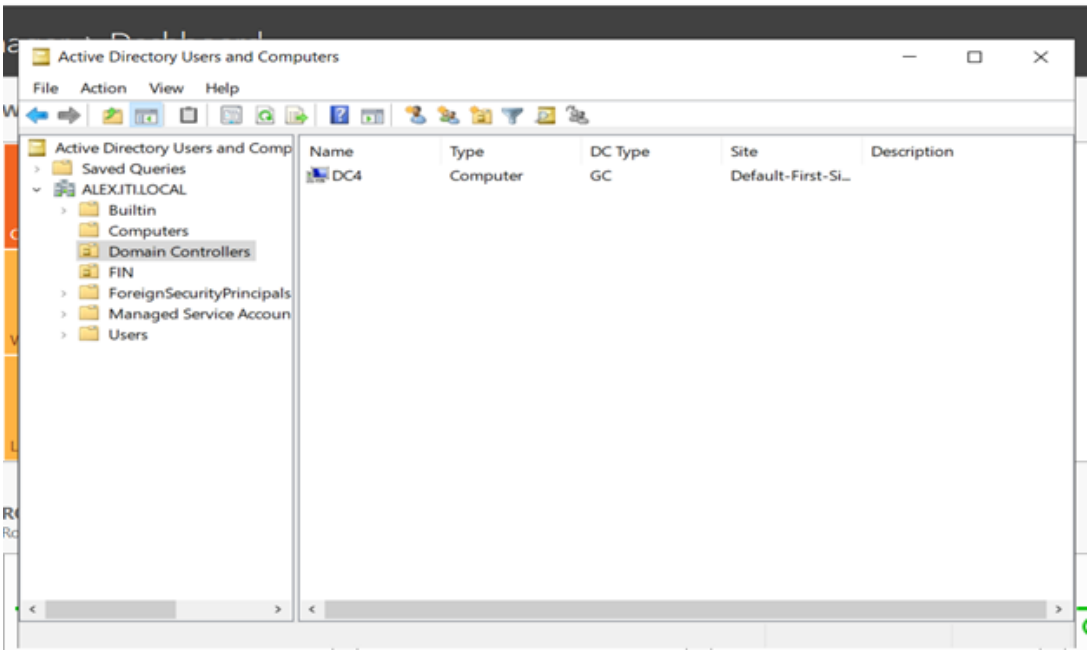


Figure 3: ALEX.ITI.LOCAL child domain structure showing DC4 and FIN organizational unit

2.3 Child Domain: ISM.ITI.LOCAL

The ISM.ITI.LOCAL child domain contains DC5 as the domain controller and organizational units including the HR-ISM department.

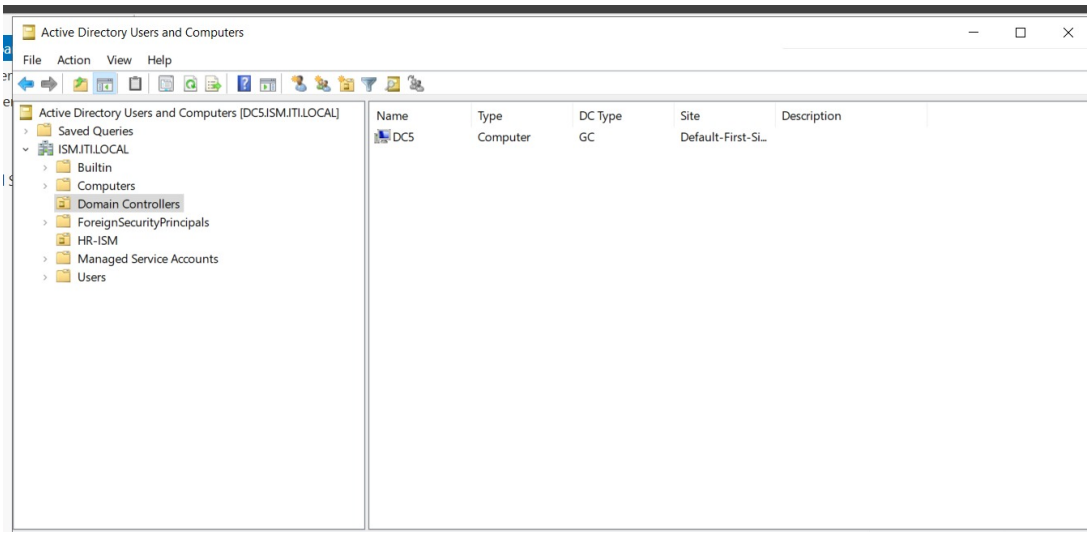


Figure 4: ISM.ITI.LOCAL child domain structure showing DC5 and HR-ISM organizational unit

3 Implementing Login Restrictions for A@ITI.local

The user account A@ITI.local was configured with two specific access restrictions: login access limited to PC1 only, and time-based restriction preventing login on Fridays.

3.1 Restricting Login to PC1 Only

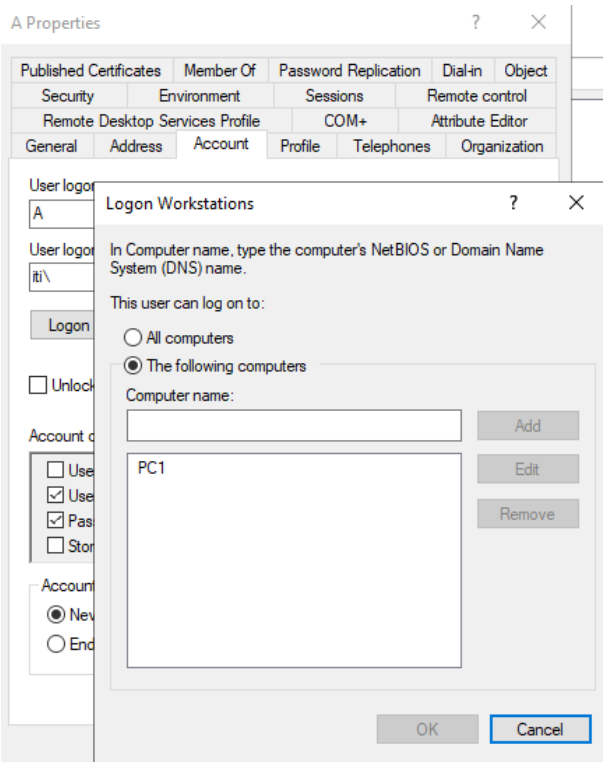


Figure 5: Logon Workstations configuration - Restricting A@ITI.local to PC1 only

3.2 Configuring Time-Based Restrictions (No Login on Fridays)

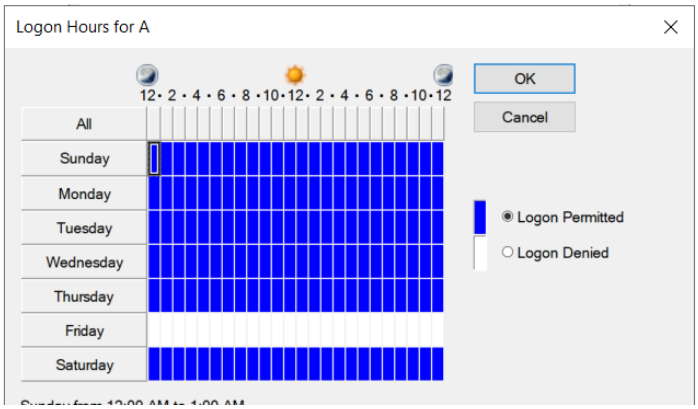


Figure 6: Logon Hours configuration - Friday is completely blocked (shown in black)

3.3 Verification

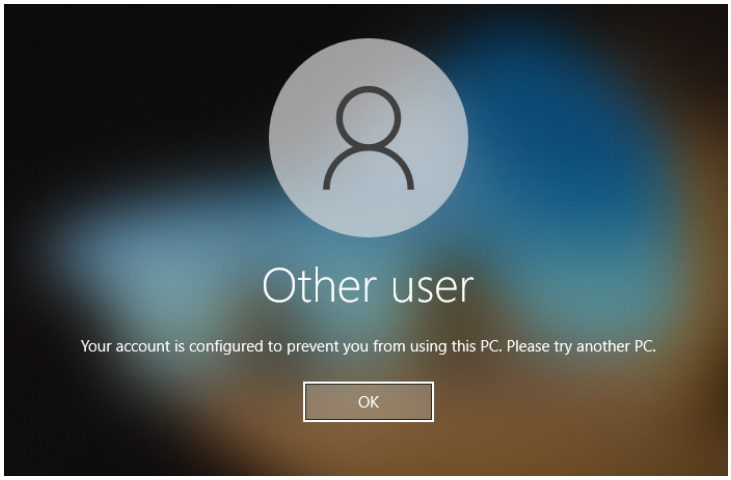


Figure 7: Error message displayed when A@ITI.local attempts to login from unauthorized computer: "Your account is configured to prevent you from using this PC. Please try another PC."

4 Configuring RODC Access for help@ITI.local

The user **help@ITI.local** was configured to login through the Read-Only Domain Controller (DC3). By default, RODCs do not store user passwords. We configured the Password Replication Policy to allow DC3 to cache help@ITI.local's password, enabling this user to authenticate even when the main domain controllers are unavailable.

4.1 Password Replication Policy Configuration

The Password Replication Policy on DC3 was configured to allow help@ITI.local's password to be stored on the RODC.

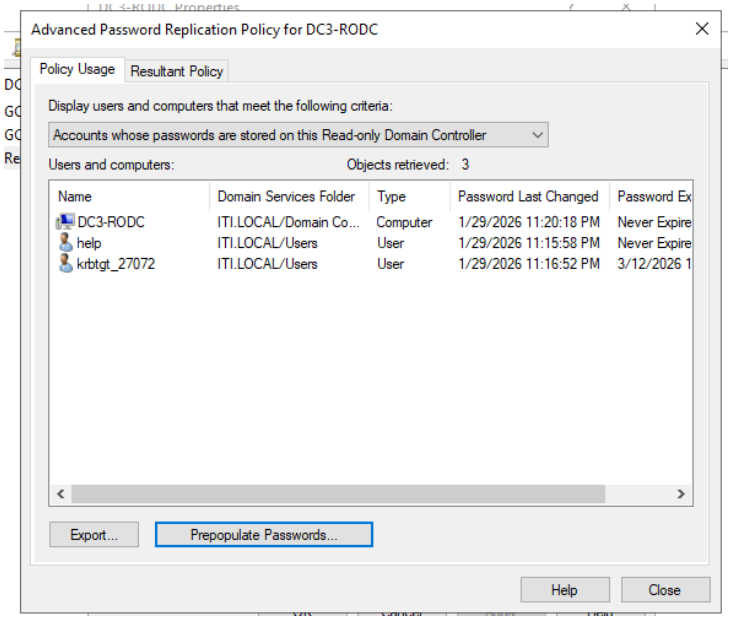


Figure 8: Password Replication Policy configuration showing help@ITI.local can replicate to RODC

4.2 Verification of RODC Authentication

The help@ITI.local user successfully logged in through DC3 (RODC) with limited permissions.

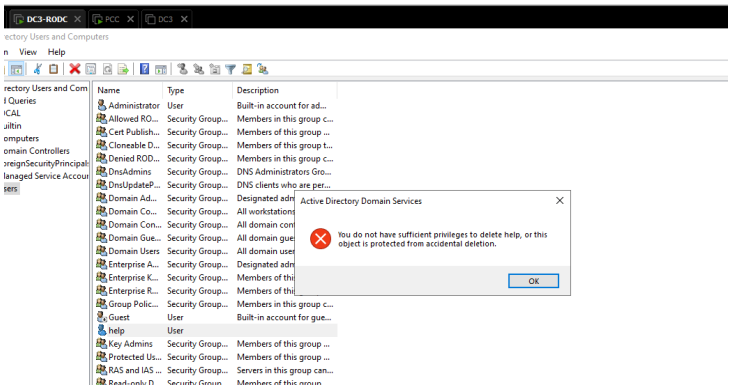


Figure 9: Verification showing help@ITI.local successfully logged in through RODC without administrative permissions

5 Implementing Restrictions for c@ITI.local

The user **c@ITI.local** was configured with multiple system restrictions to demonstrate Group Policy security controls. Three specific restrictions were applied: blocking access to Control Panel, preventing use of removable storage devices (Flash memory), and enforcing a mandatory desktop wallpaper with the ITI logo.

5.1 Configuration

5.1.1 Removing Access to Control Panel

Group Policy was configured to prevent c@ITI.local from accessing the Control Panel, restricting the user's ability to modify system settings.

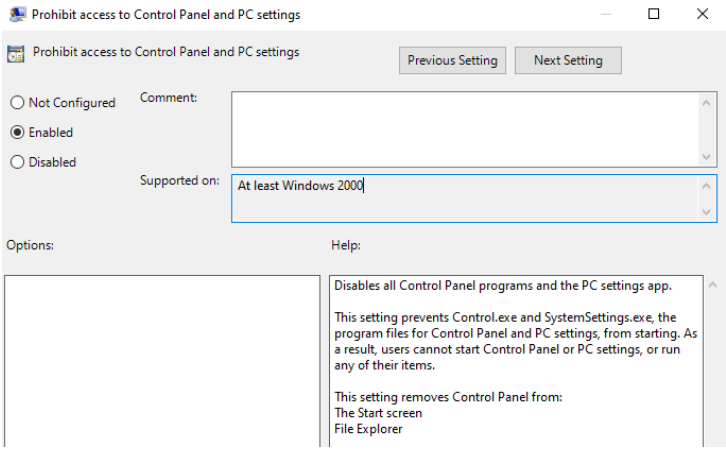


Figure 10: GPO configuration to disable Control Panel access for c@ITI.local

5.1.2 Blocking Removable Storage and Flash Memory Access

Group Policy was configured to block all removable storage devices including USB flash drives, preventing c@ITI.local from accessing or using any external storage media.

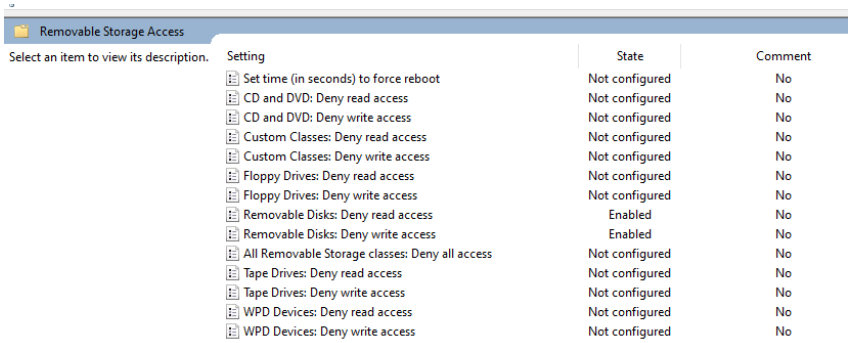


Figure 11: GPO configuration to block removable storage access for c@ITI.local

5.1.3 Enforcing Desktop Wallpaper (ITI Logo)

Group Policy was configured to enforce a mandatory desktop wallpaper displaying the ITI logo, preventing c@ITI.local from changing the desktop background.

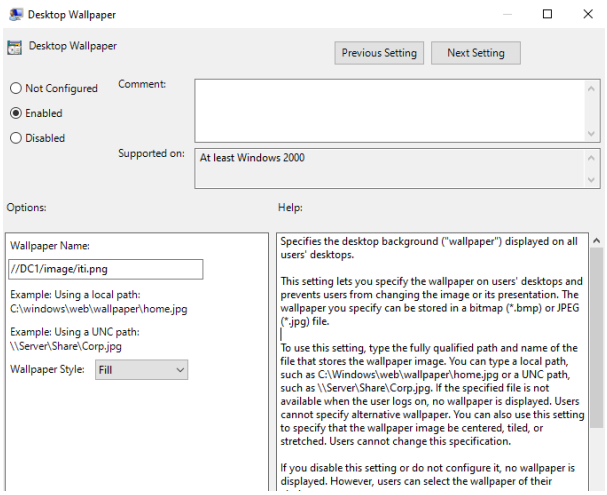


Figure 12: GPO configuration to enforce ITI logo wallpaper for c@ITI.local

5.2 Verification

5.2.1 Verifying Control Panel Restriction

When c@ITI.local attempts to access Control Panel, the action is blocked by Group Policy.

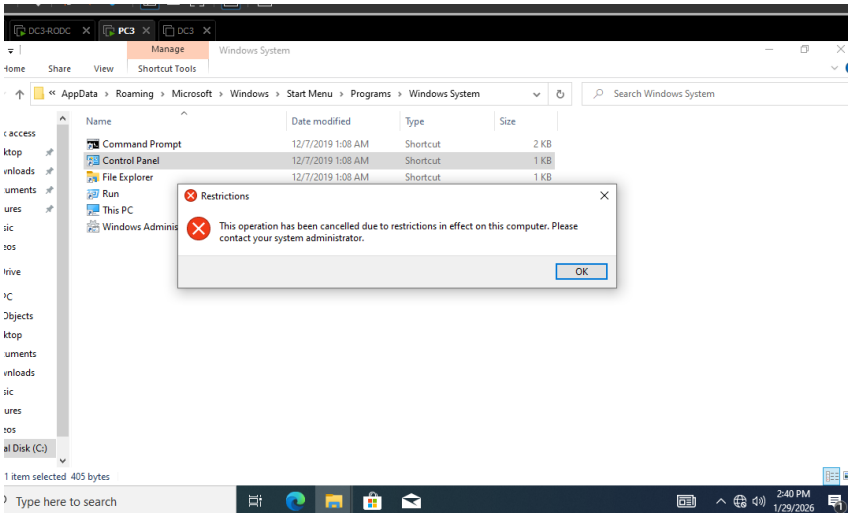


Figure 13: Verification showing Control Panel access is denied for c@ITI.local

5.2.2 Verifying Removable Storage Restriction

When c@ITI.local attempts to access removable storage devices (USB flash drives), the access is blocked by Group Policy.

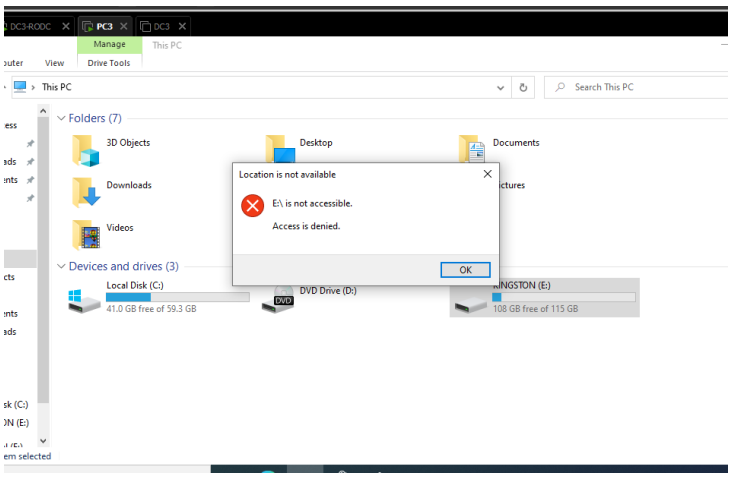


Figure 14: Verification showing removable storage access is denied for c@ITI.local

5.2.3 Verifying Desktop Wallpaper Enforcement

The desktop wallpaper is automatically set to the ITI logo and c@ITI.local cannot change it.

Figure 15: Verification showing ITI logo wallpaper is enforced for c@ITI.local

6 Software Deployment Using Group Policy Objects (GPO)

Group Policy Objects can be used to automatically deploy software applications across domain computers. In this task, VLC Media Player was deployed to PC2 using GPO, demonstrating centralized software management capabilities in Active Directory environments.

6.1 GPO Configuration for VLC Installation

6.1.1 Creating Software Deployment GPO

A new Group Policy Object was created to deploy VLC Media Player. The VLC installation package (MSI file) was added to the GPO software installation settings.

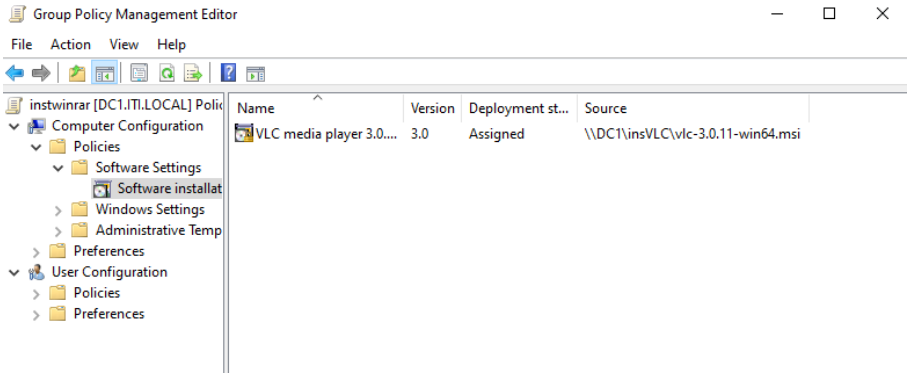
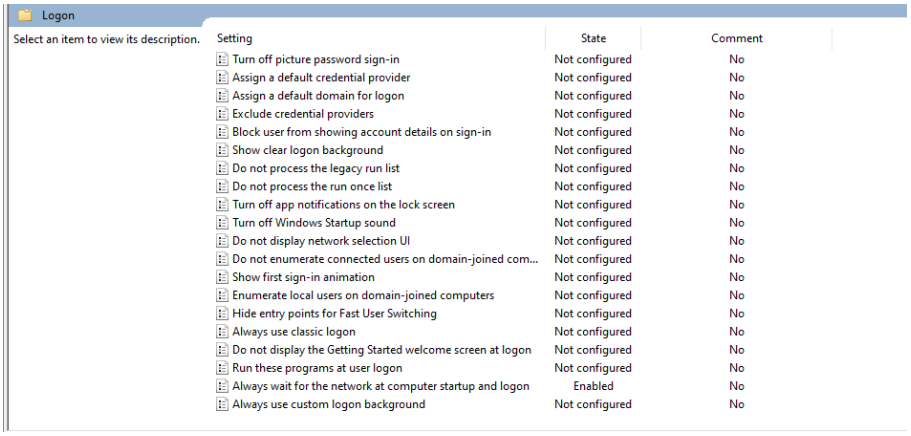


Figure 16: GPO Software Installation configuration for VLC Media Player

6.1.2 Configuring Startup Policy

The GPO was configured to wait for network at startup and apply software installation during computer startup to ensure proper deployment.



Logon

Select an item to view its description.

Setting	State	Comment
Turn off picture password sign-in	Not configured	No
Assign a default credential provider	Not configured	No
Assign a default domain for logon	Not configured	No
Exclude credential providers	Not configured	No
Block user from showing account details on sign-in	Not configured	No
Show clear logon background	Not configured	No
Do not process the legacy run list	Not configured	No
Do not process the run once list	Not configured	No
Turn off app notifications on the lock screen	Not configured	No
Turn off Windows Startup sound	Not configured	No
Do not display network selection UI	Not configured	No
Do not enumerate connected users on domain-joined com...	Not configured	No
Show first sign-in animation	Not configured	No
Enumerate local users on domain-joined computers	Not configured	No
Hide entry points for Fast User Switching	Not configured	No
Always use classic logon	Not configured	No
Do not display the Getting Started welcome screen at logon	Not configured	No
Run these programs at user logon	Not configured	No
Always wait for the network at computer startup and logon	Enabled	No
Always use custom logon background	Not configured	No

Figure 17: GPO configuration - Always wait for network at computer startup and logon

The GPO was then linked to the appropriate Organizational Unit containing PC2 to apply the policy.

6.2 Verification of VLC Installation

After applying the GPO and restarting PC2, VLC Media Player was automatically installed on the system.

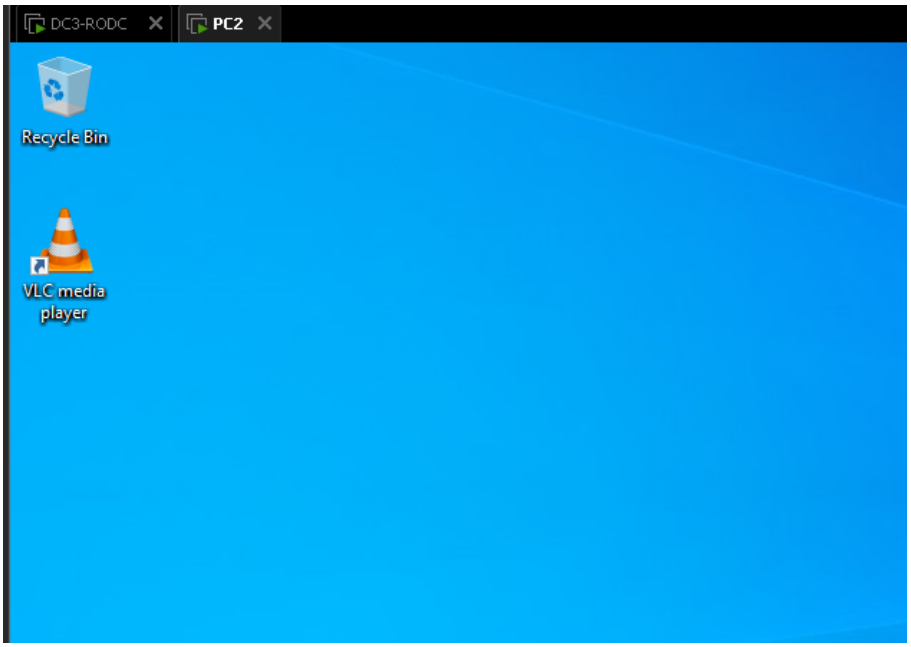


Figure 18: Verification showing VLC Media Player successfully installed on PC2 via GPO

7 Administrative Delegation for B@iti.local

Administrative delegation allows specific users to perform limited administrative tasks without granting full Domain Administrator privileges. The user **B@iti.local** was granted the ability to remotely login to DC1 using Remote Desktop Protocol (RDP) without being a member of the Administrators group, demonstrating the principle of least privilege.

7.1 Configuration

7.1.1 Adding B@iti.local to Remote Desktop Users Group

The user B@iti.local was added to the Remote Desktop Users group on DC1, which grants the permission to connect remotely via RDP without administrative privileges.

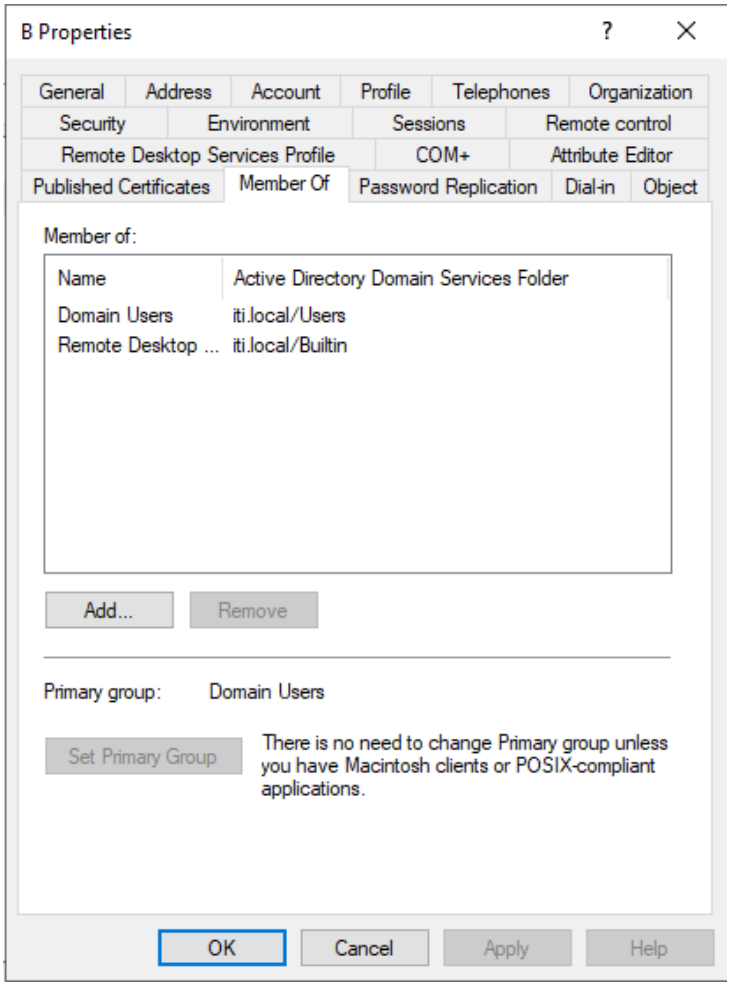


Figure 19: Adding B@iti.local to Remote Desktop Users group on DC1

7.1.2 Enabling Remote Desktop via Group Policy

Group Policy was configured to allow Remote Desktop connections and specifically grant B@iti.local the right to login remotely to DC1.

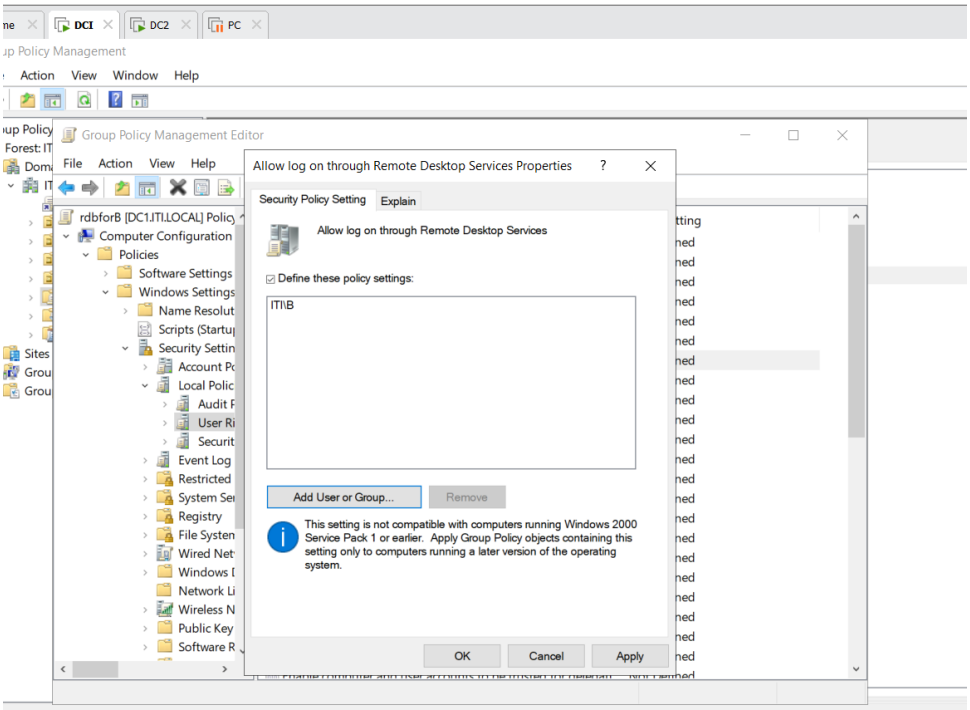


Figure 20: GPO configuration allowing B@iti.local remote desktop access to DC1

7.2 Verification of Delegated Access

B@iti.local successfully connected to DC1 via Remote Desktop without being a member of the Domain Administrators or local Administrators group.

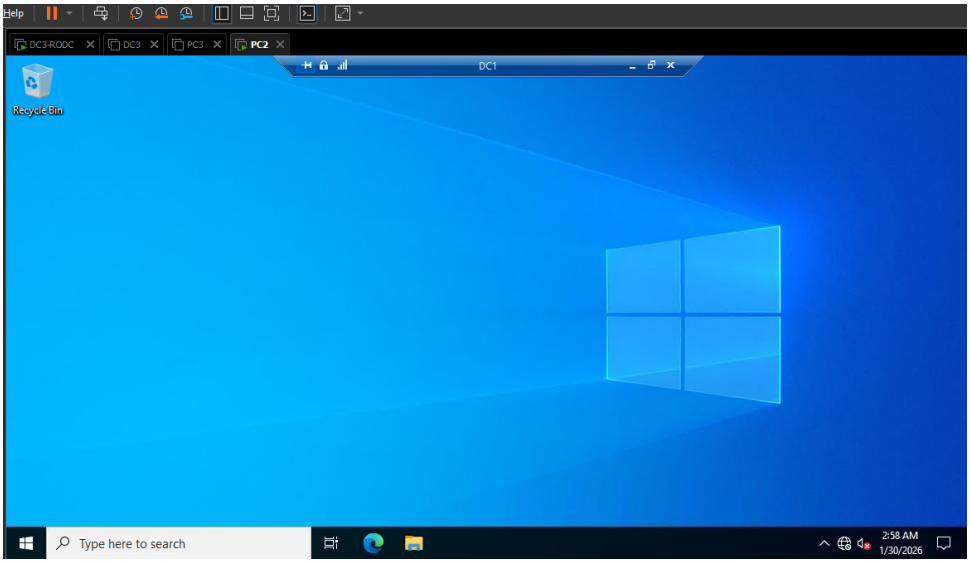


Figure 21: Verification showing B@iti.local successfully logged into DC1 remotely without administrator privileges

8 Roaming User Profiles for A@Ism.ITI.Local

Roaming user profiles allow users to have the same desktop environment, settings, and documents regardless of which computer they login to. The user **A@Ism.ITI.Local** was configured

with a roaming profile to maintain consistent experience across multiple workstations (PC5, PC1, and PC4).

8.1 Configuration

8.1.1 Creating Shared Folder for Profiles

A shared folder was created on the server to store roaming user profiles. This folder must be accessible by domain users with appropriate permissions.

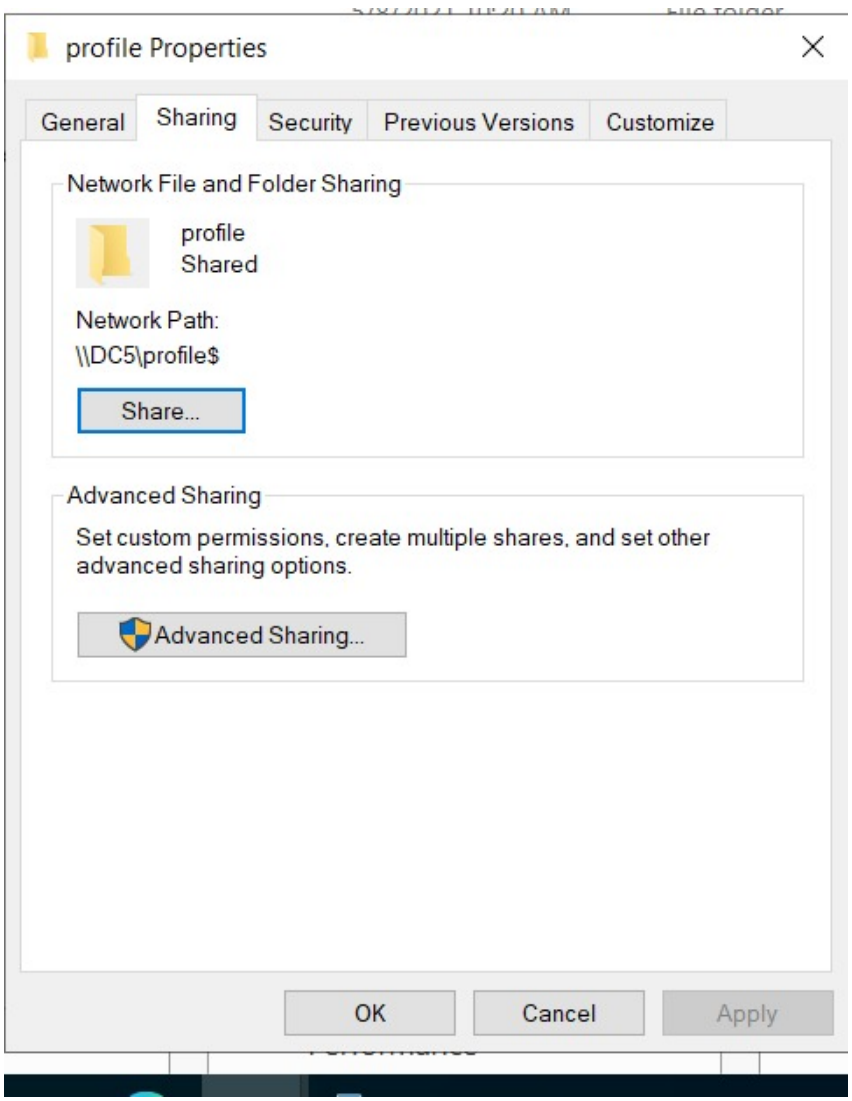


Figure 22: Creating and configuring shared folder for roaming profiles in C: drive

8.1.2 Configuring User Profile Path

The user account properties for A@Ism.ITI.Local were configured to point to the shared profile folder. The profile path was set in Active Directory Users and Computers.

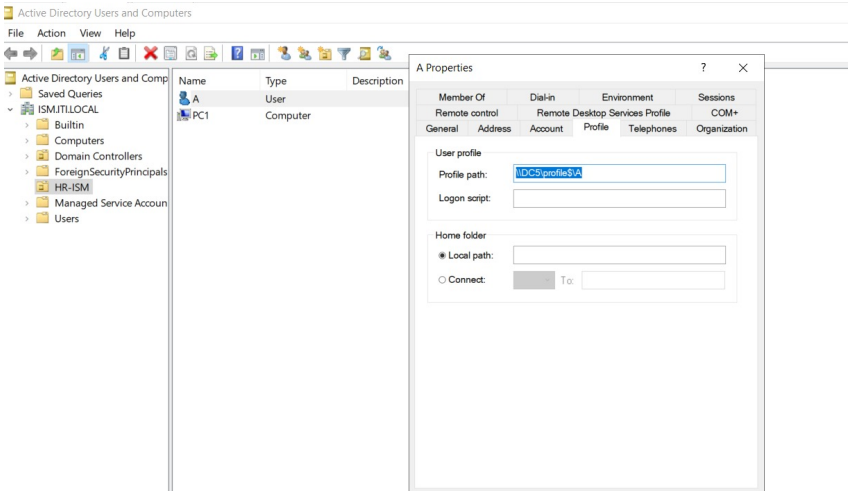


Figure 23: Setting roaming profile path for A@Ism.ITI.Local in user properties

8.2 Verification of Roaming Profile

A@Ism.ITI.Local successfully logged in to multiple computers (PC5, PC1, and PC4) with the same profile settings, documents, and desktop environment synchronized across all workstations.

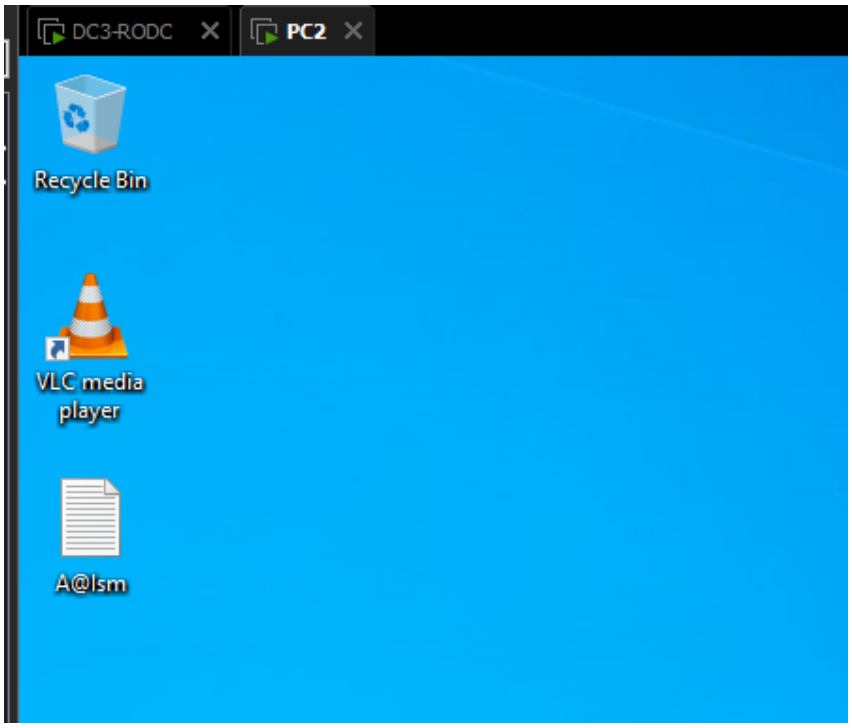


Figure 24: Verification showing A@Ism.ITI.Local using the same roaming profile across PC5, PC1, and PC4

9 Remote Server Configuration (External to Domain)

9.1 Overview

This section describes the configuration of a remote server located outside the ITI.Local domain. The server provides multiple network services including web hosting, DNS, DHCP, Remote Desktop access, and FTP file transfer capabilities.

The implementation includes:

- Two websites (web1.com and web2.com) hosted on IIS Web Server
- DNS service for local name resolution
- Secondary DNS zone integration with the primary domain controller
- DHCP service providing dynamic IP addresses to PC6
- Remote Desktop Protocol (RDP) access for remote management
- FTP service for website file management
- Shared folder with user-specific permissions

A local user named **RemoteUser** was created to manage the server remotely from PC6 using RDP and FTP protocols.

9.2 Web Server Configuration

IIS (Internet Information Services) was installed on the remote server to host two separate websites accessible through different hostnames. The Web Server (IIS) role was installed on the remote server to provide web hosting capabilities. Two websites were created: web1.com and web2.com, each configured to respond to their respective hostname requests. Both websites are accessible through hostname-based routing on the same server IP address (10.166.108.47).

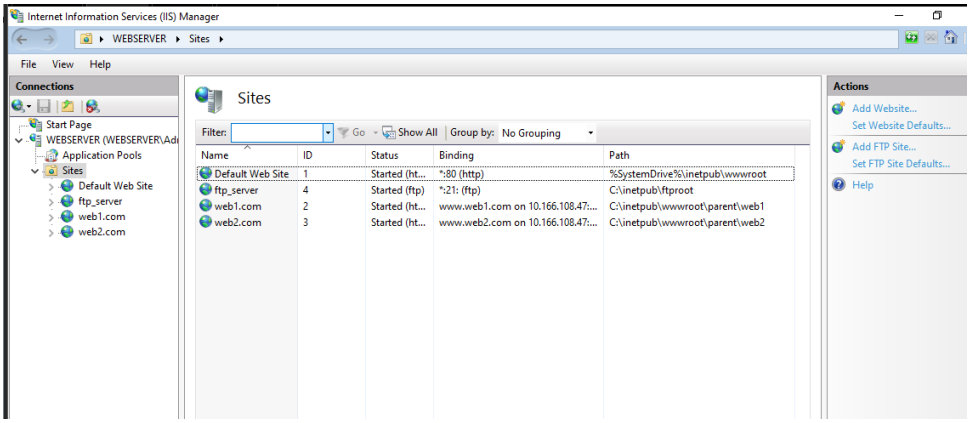


Figure 25: Configuration of websites in IIS

9.3 DNS Configuration

Two different methods were implemented to allow users to access the websites through DNS name resolution.

9.3.1 Method 1: Local DNS Service on Remote Server

DNS Server role was installed on the remote server to provide local name resolution for the hosted websites.
Creating DNS Records: Forward lookup zones were created for both websites, with A records pointing to the server's IP address (10.166.108.47).

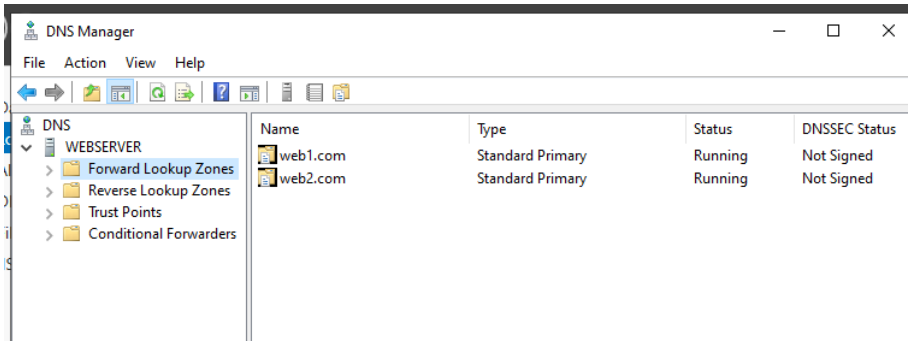


Figure 26: DNS A record for websites on local DNS server

9.3.2 Method 2: Secondary Zone on Primary Domain Controller

To allow domain users in ITI.Local to access these websites, a secondary DNS zone was configured on DC1 (Primary Domain Controller) that refers to the remote server's DNS service.

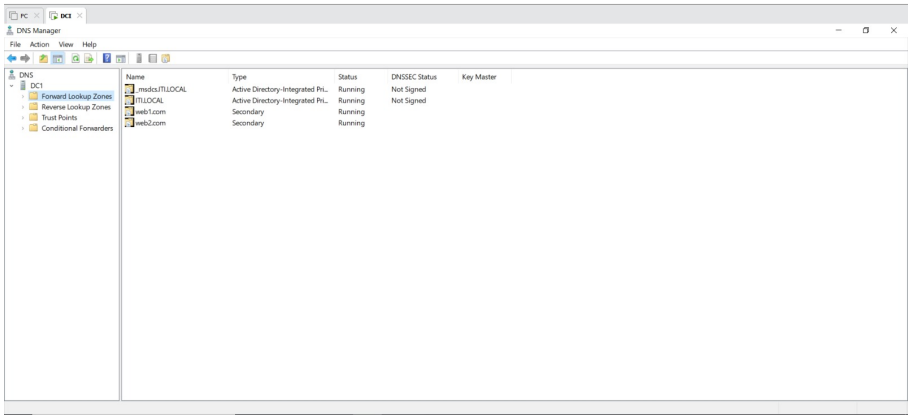


Figure 27: Secondary zone configuration on DC1 for websites

9.3.3 Verification of DNS Configuration and Secondary Zone

Testing Website Access: Domain users successfully accessed the websites through their web browsers, confirming proper DNS resolution.

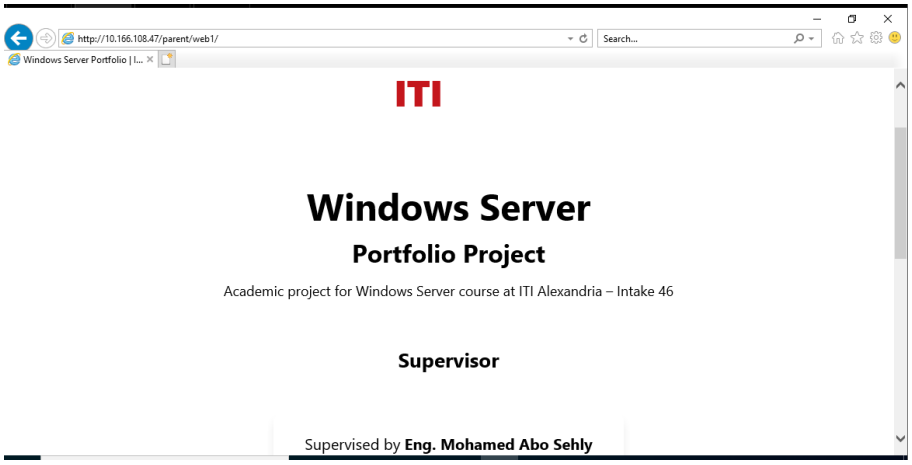


Figure 28: Successful access to website from domain computer on website 1

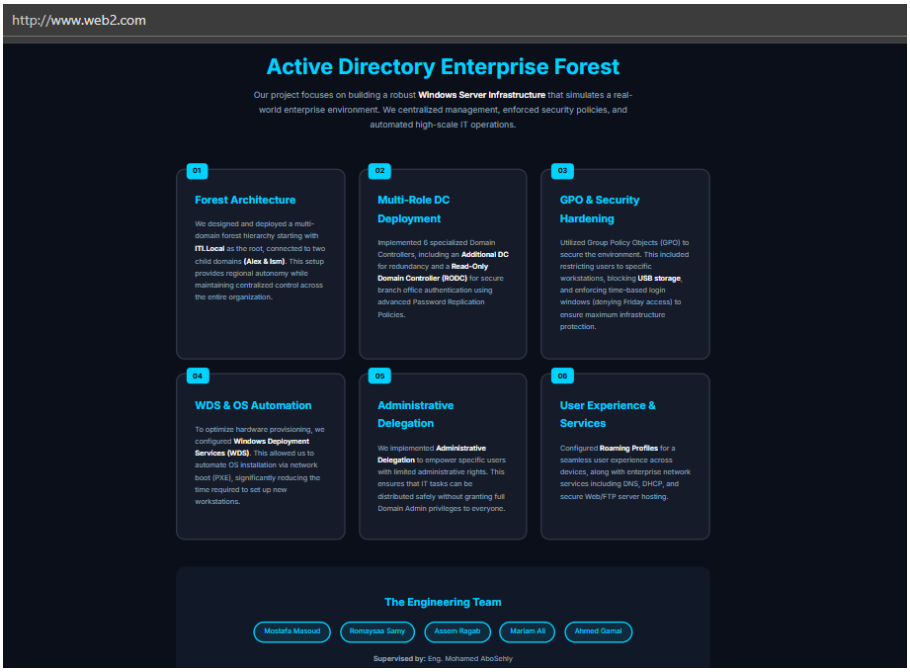


Figure 29: Successful access to website from domain computer on website 2

Testing DNS Resolution with nslookup: The nslookup command was executed from a PC within the ITI.Local parent domain to verify DNS resolution. The response shows an **authoritative answer** (no "non-authoritative" message appears), confirming that DC1 is responding authoritatively from its secondary zone. This proves the secondary zone configuration is working correctly and the zone transfer from the remote DNS server was successful.

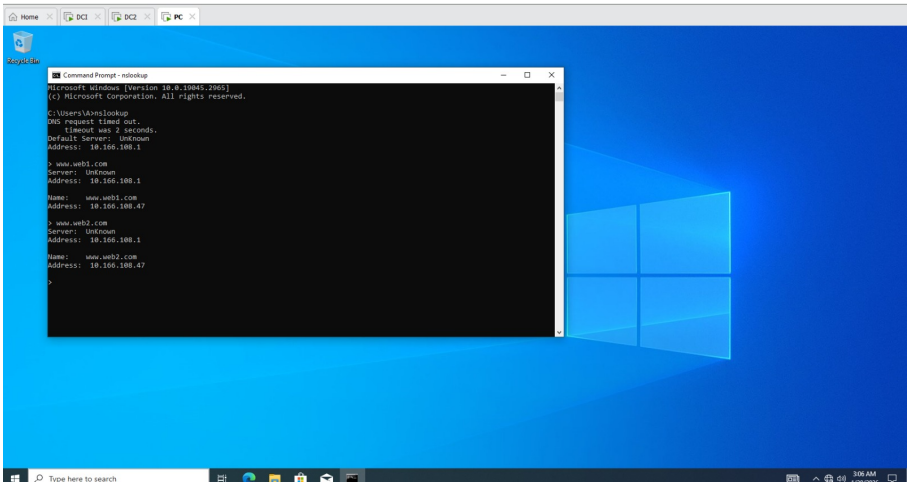


Figure 30: nslookup from domain PC showing authoritative response, confirming secondary zone is functioning properly

9.4 Installing DHCP Server Role

The DHCP Server role was installed on the remote server.

9.4.1 Creating DHCP Scope

A DHCP scope was configured to assign IP addresses dynamically from the 10.166.108.x network range.

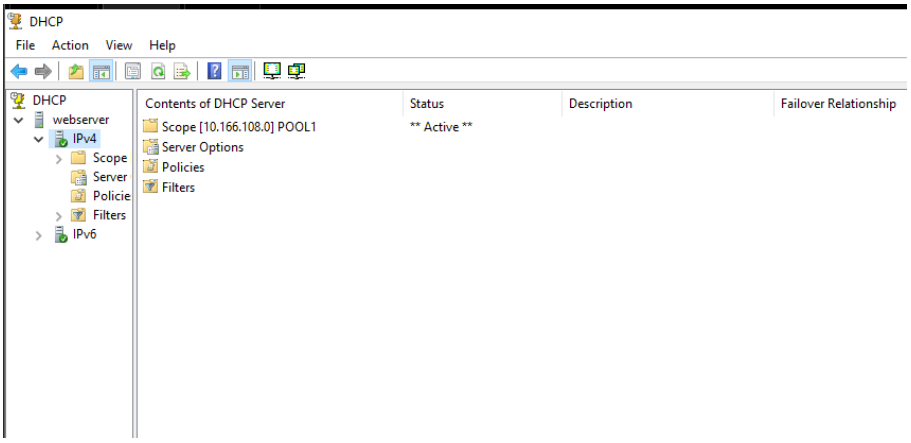


Figure 31: DHCP scope configuration for dynamic IP assignment

9.4.2 Verification of DHCP Configuration

PC6 successfully obtained an IP address (10.166.108.50) automatically from the DHCP server, confirming proper DHCP service configuration and functionality.

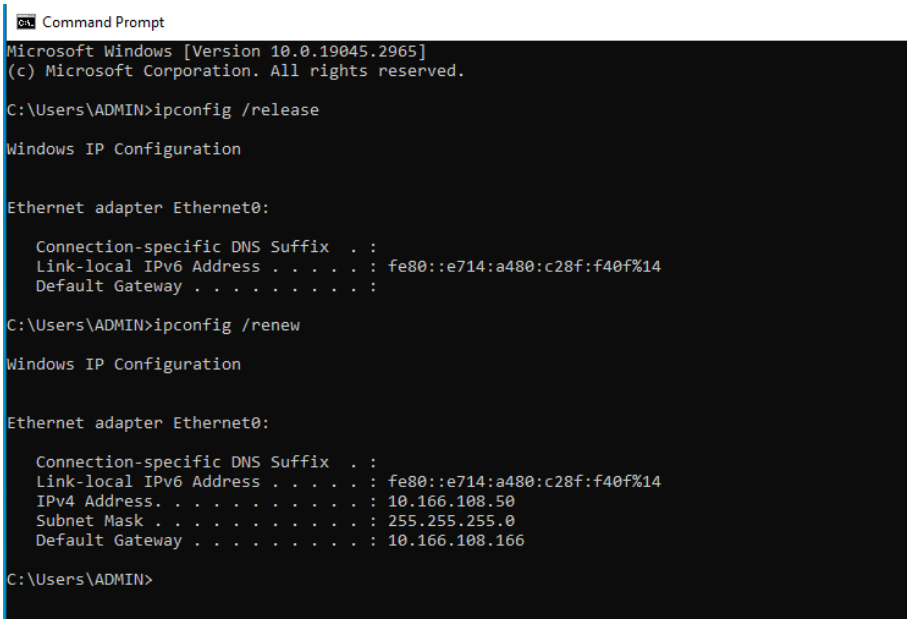


Figure 32: PC6 successfully obtained IP address 10.166.108.50 from DHCP server

9.5 Remote Desktop Protocol (RDP) Configuration

A local user account named **RemoteUser** was created on the remote server with administrative privileges.

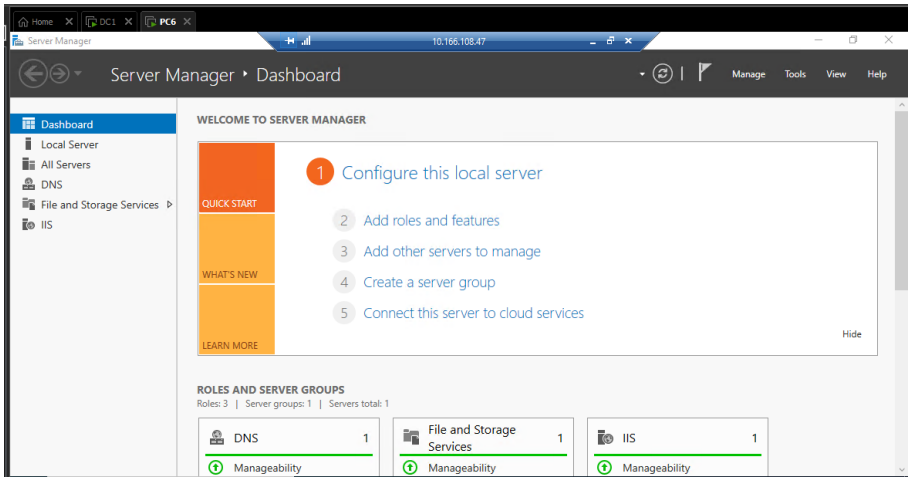


Figure 33: RemoteUser successfully connected to remote server via RDP from PC6

9.6 FTP Server Configuration

FTP (File Transfer Protocol) service was installed to allow RemoteUser to manage website files remotely.

9.6.1 Verification of FTP Access

RemoteUser successfully connected to the FTP server from PC6 and accessed the web1.com files with read and write permissions.

Figure 34: FTP connection verification showing user mostafa successfully connecting to FTP server (10.166.108.47), logging in, and retrieving files from web1.com

9.6.2 Verification: User Can Create Own Folder

User Mostafa successfully created a folder within the shared directory and has full access to work within it.

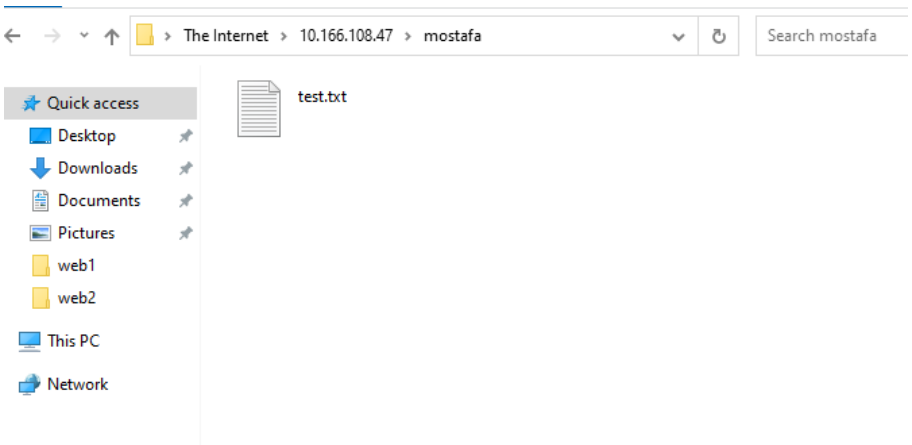


Figure 35: User Mostafa creating and accessing his own folder

9.6.3 Verification: User Cannot Access Other Users' Folders

User Mostafa attempted to access a folder named "test" created by another user, and access was denied.

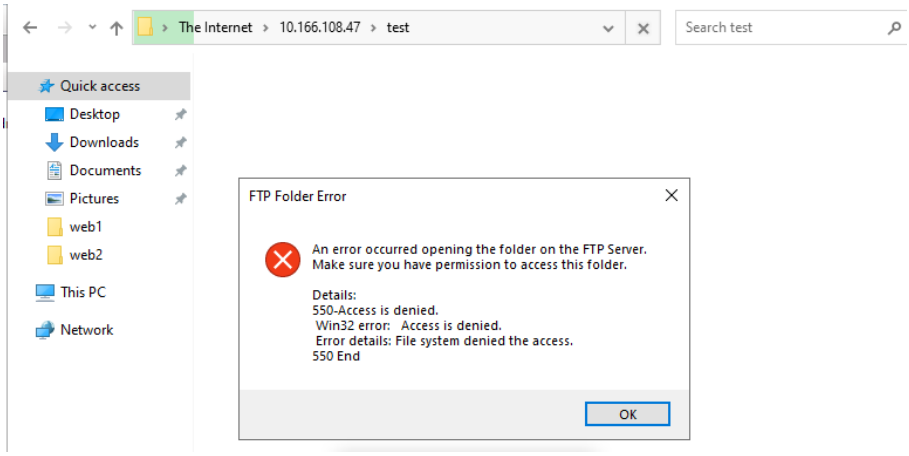


Figure 36: User Mostafa's access denied to another user's folder (test)

10 Windows Deployment Services (WDS) for Automated Deployment

Windows Deployment Services (WDS) was configured on DC6 to automate the deployment of 50 new computers and create 50 user accounts in the domain. This demonstrates enterprise-scale automated provisioning capabilities.

10.1 Creating 50 User Accounts Using Batch Script

Before deploying workstations, 50 user accounts were created in Active Directory using a CMD batch script to automate the bulk user creation process.

10.1.1 CMD Batch Script for Bulk User Creation

A CMD batch script was developed to automatically create 50 user accounts with standardized naming convention and attributes using the `dsadd user` command.

```
@echo off
set DOMAIN=dc=iti,dc=local
set OU=ou=sales

for /L %%i in (1,1,50) do (
 dsadd user "cn=user%%i,%OU%,%DOMAIN%" ^
  -fn test%%i ^
  -ln user%%i ^
  -samid test_user%%i ^
  -upn test_user%%i@iti.local ^
  -pwd P@ssw0rd123 ^
  -mustchpwd yes ^
  -disabled no
)

echo Done creating 50 users
pause
```

Figure 37: CMD Batch script for creating 50 user accounts automatically in Sales OU

The script creates users named user1 through user50 in the Sales organizational unit (OU=sales) with the following attributes:

- User Principal Name: test_user%i@iti.local
- Password: P@ssw0rd123
- Password never expires
- Account enabled

10.1.2 Verification of User Creation

The script was executed successfully, and all 50 user accounts were created in Active Directory under the Sales OU.

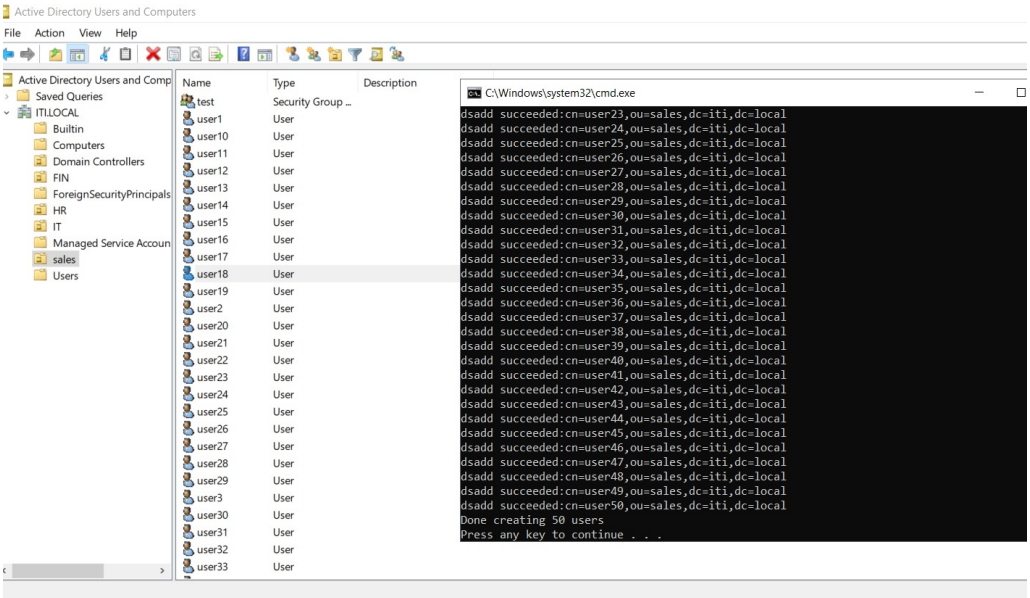


Figure 38: Verification showing 50 users successfully created in Active Directory

10.2 WDS Server Configuration on DC6

10.2.1 Installing WDS Server Role

The Windows Deployment Services role was installed on DC6 to enable network-based operating system deployment.

10.2.2 Configuring DHCP Scope for WDS

A dedicated DHCP scope was created to provide IP addresses for computers during PXE boot and automated deployment.

DHCP Address Pool Configuration:

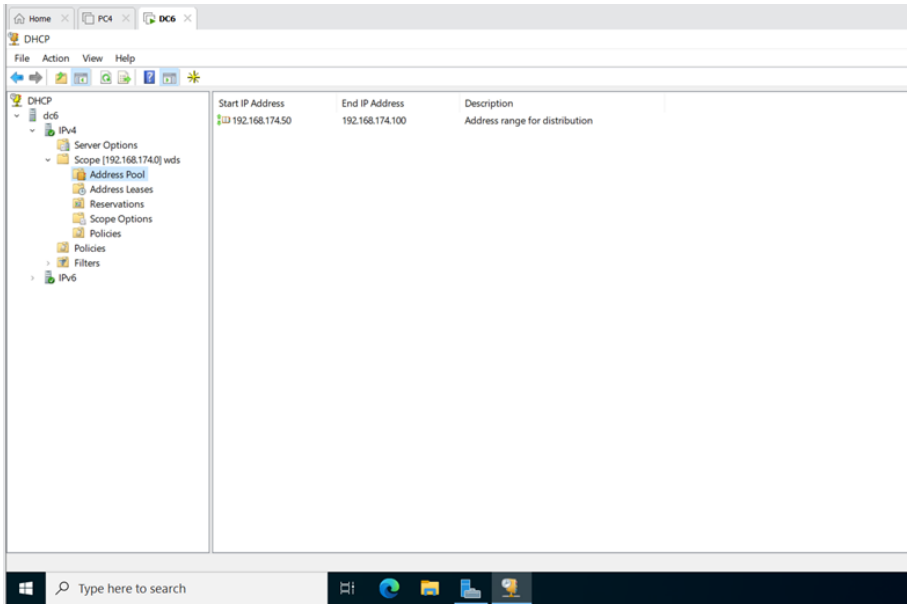


Figure 39: DHCP address pool configuration for WDS automated deployment

DHCP Scope Options Configuration:

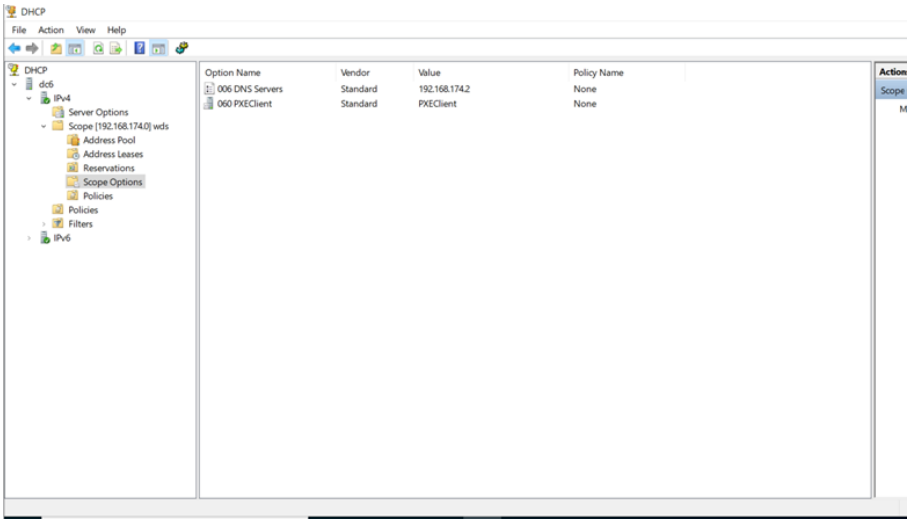


Figure 40: DHCP scope options for WDS including gateway and DNS settings

10.2.3 Adding Boot Image (boot.wim)

The boot image (boot.wim) was added to WDS to provide the Windows Pre-installation Environment (WinPE) that clients boot into during network installation.

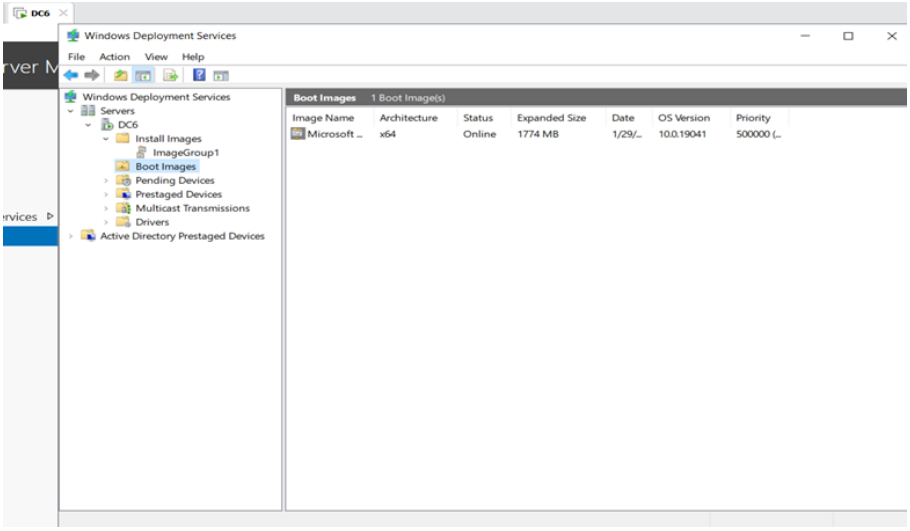


Figure 41: WDS boot image (boot.wim) configuration for PXE boot environment

10.2.4 Adding Install Image (install.wim)

The installation image (install.wim) was added to WDS, containing the Windows operating system that will be deployed to client computers.

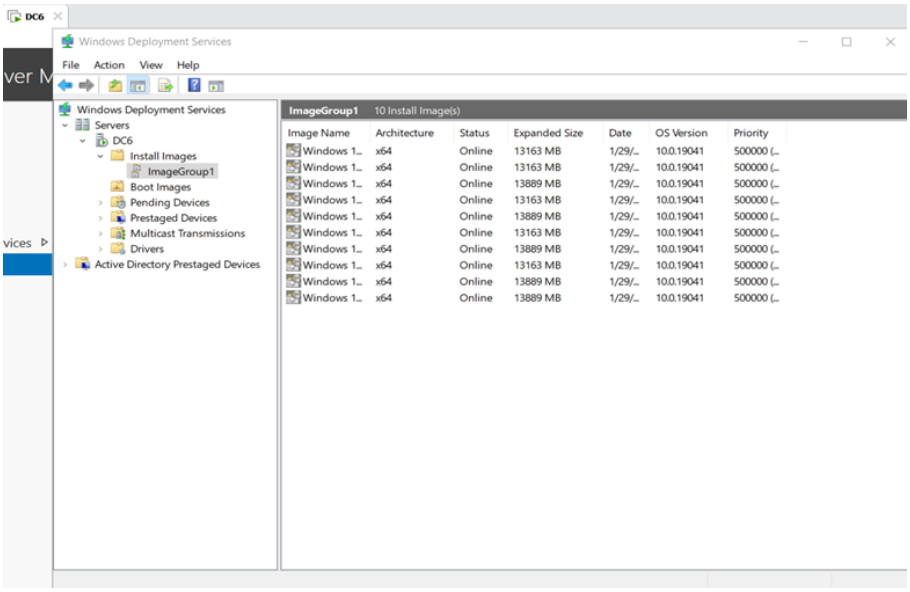


Figure 42: WDS install image (install.wim) configuration for operating system deployment

10.3 Testing WDS Deployment

The WDS configuration was tested by deploying Windows to a test computer using PXE network boot.

10.3.1 PXE Boot Process

A test computer was configured to boot from the network, connecting to the WDS server.

Figure 43: Test computer initiating PXE boot and connecting to WDS server

10.3.2 Windows Installation via WDS

The Windows installation process began automatically over the network, receiving the operating system from the WDS server.

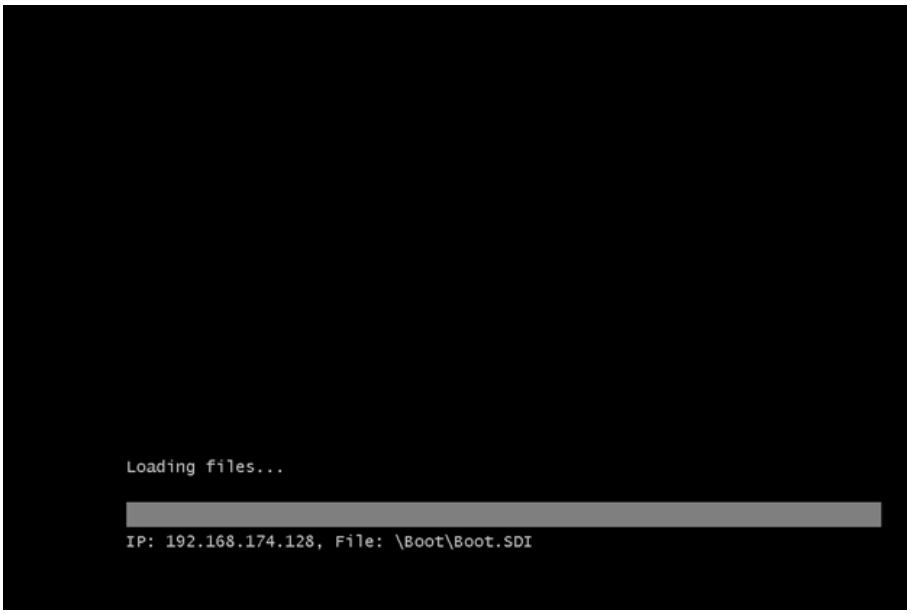


Figure 44: Automated Windows installation in progress from WDS server

10.3.3 Successful Deployment Verification

The deployment completed successfully, with Windows installed and the computer ready to join the domain.

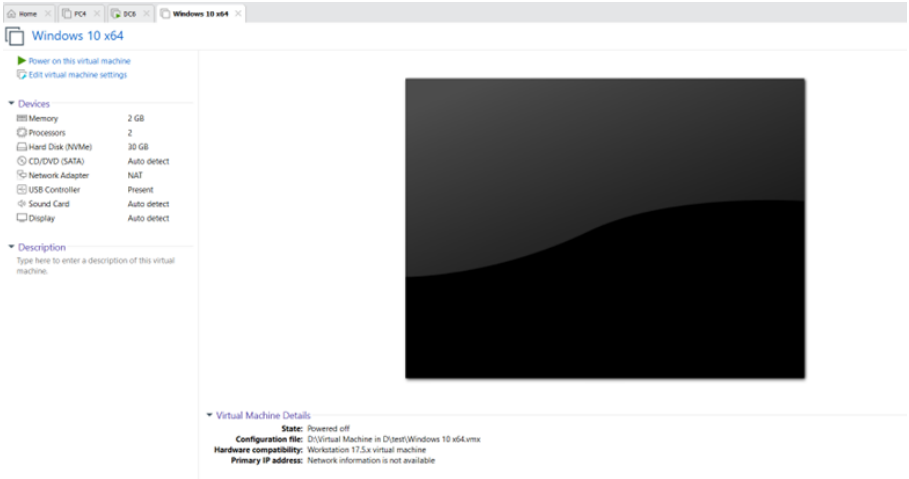


Figure 45: Successful Windows deployment completion via WDS

11 Conclusion

This project successfully implemented a complete Active Directory infrastructure with ITI.Local forest, two child domains, six domain controllers, and various services including Group Policy, roaming profiles, WDS, and external network services.

We express our sincere gratitude to **ENG. Mohamed AboSehly** for his exceptional guidance, continuous support, and expert supervision throughout this project. His extensive knowledge and dedication were instrumental in our success and professional development.

Thank you for being an outstanding mentor.